

for which, using Eqs. (27.4.19) and (27.4.20),

$$\begin{aligned}(M^\dagger M)_{nm} &= (\mathcal{M}^\dagger \mathcal{M})_{nm} + 2 \sum_A (t_A \phi_0)_n (t_A \phi_0)_m^*, \\ (M^\dagger M)_{nA} &= (M^\dagger M)_{An} = 0, \\ (M^\dagger M)_{AB} &= 2(\phi_0^\dagger t_B t_A \phi_0) = (\phi_0^\dagger \{t_B, t_A\} \phi_0).\end{aligned}\tag{27.4.31}$$

The eigenvectors of the matrix (27.4.30) are of three types. First are those of the form

$$z = \begin{bmatrix} \zeta \\ 0 \end{bmatrix}, \tag{27.4.32}$$

with eigenvalues m^2 , where ζ_n and m^2 are any eigenvectors and corresponding eigenvalues of $\mathcal{M}^\dagger \mathcal{M}$. Next are those of the form

$$g = \begin{bmatrix} 0 \\ c \end{bmatrix}, \tag{27.4.33}$$

with eigenvalues μ^2 , where c_B and μ^2 are any eigenvectors and eigenvalues of the matrix $(\phi_0^\dagger \{t_B, t_A\} \phi_0)$. Finally, there are those of the form

$$h = \begin{bmatrix} \sum_B c_B t_B \phi_0 \\ 0 \end{bmatrix}, \tag{27.4.34}$$

with eigenvalues μ^2 , where c_B and μ^2 are again any eigenvectors and eigenvalues of the matrix $(\phi_0^\dagger \{t_B, t_A\} \phi_0)$. The only exception is that the eigenvectors c of this matrix with eigenvalue zero have $\sum_A c_A t_A \phi_0 = 0$, corresponding to unbroken symmetries, so that in this case the vector (27.4.34) vanishes and we have only the eigenvector (27.4.33). Thus there is one Majorana fermion of each mass m satisfying Eq. (27.4.26), two Majorana fermions of each non-zero mass μ satisfying Eq. (27.4.22), and one Majorana fermion of zero mass for each unbroken gauge symmetry.

Spin 1

The mass terms in the Lagrangian for the gauge fields arise from the part of the first term in Eq. (27.4.1) that is of second order in the gauge field V_A^μ :

$$\mathcal{L}_V = - \sum_{nAB} (t_A \phi_0)_n^* (t_B \phi_0)_n V_{A\mu} V_B^\mu. \tag{27.4.35}$$

Since the fields $V_{A\mu}$ are real, their mass-squared matrix is the matrix in Eq. (27.4.23):

$$(\mu^2)_{AB} = (\phi_0^\dagger \{t_B, t_A\} \phi_0). \tag{27.4.36}$$